

Support-Controlled Reversion and the Exact One-Exponential Substitution Group

Dense Hahn exponents, Newton denominator stability,
Cayley sector coefficients, and a convergent mixed example

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Abstract

We address three explicitly stated questions in the ProveIt volume on polynomial–logarithmic transseries. First, dense real exponent groups do not require a new index set for Lagrange–Bürmann reversion: the ordinary positive-integer sum remains valid, with a precise valuation bound and a support certificate. Second, every Newton iterate stays in the input Puiseux lattice; more strongly, its normalized support remains in the positive monoid generated by the normalized input support. Third, adjoining a single exponential does not preserve the entire polynomial–logarithmic substitution group. We give an exact obstruction and a replacement theorem. In $\mathbb{R}[L]((t^\Gamma))[[q]]$, with $t = X^{-1}$, $L = \log X$ and $q = e^{-\mu X}$, a map $X + P(L) + U + h$, where $v(U) > 0$ and $h \in q\mathcal{E}_{\Gamma, \mu}$, admits the normalized differential-compatible substitution precisely when $P(L) = aL + b$ and $\mu a \in \Gamma$. This class is maximal among maps with a sublinear Hahn base displacement. The admissible maps form a group, and inversion is given by an explicitly summable formula after the zero-sector core is inverted exactly. For a single exponential perturbation we determine the leading block of every inverse sector: its coefficient is governed by $n^{n-1}/n!$, with exact exponent and logarithmic degree. A worked example, $X + \log X + e^{-X}$, has rational sector coefficients, a sharp pole hierarchy, and a uniform geometric remainder bound on a positive real ray.

All central assertions are proved here. These are resolutions of specified repository questions and explicit extensions of its calculus, not a claim that the classical theory of transseries reversion is new. In particular, formal exponential-sector closure is not Borel summability or resurgence. The accompanying finite symbolic checks are not a Lean formalization.

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1 Research target, provenance, and scope

1.1 The questions being answered

The canonical ProveIt source is *Transseries: the polynomial–logarithmic calculus, series reversal at infinity, and the inversion of rapidly growing functions*, in the `Transseries_And_Inversion` package under the Fabius research frontiers [1, 2]. Its extension chapter asks for a Lagrange–Bürmann formula at dense exponents, a bound on Puiseux denominators in Newton iteration, and a one-exponential enlargement with workable composition and reversion. We use the source’s stable labels, rather than page numbers, to specify the targets:

Repository question	Result of this article
<code>plt:rmk:ext-open-burmann</code>	Integer-indexed formula on the full Hahn class, with an exact finite index bound for each coefficient; Section 3.
<code>plt:rmk:ext-open-denominators</code>	$\sigma(s) = s$ suffices and is minimal under divisibility; a stronger support-monoid statement holds; Section 4.
<code>plt:rmk:ext-open-resurgence</code>	A negative answer to unrestricted “verbatim” closure, an exact admissibility criterion, and a closed formal sector calculus; Sections 5–7.

The last row is deliberately qualified. The repository question includes analytic language. We settle its formal algebraic component and identify an obstruction to the proposed unrestricted substitution class. We do not construct a universal resurgent completion or prove analytic realizability for arbitrary formal inputs.

The source snapshot inspected through the canonical raw-source address and the package README was retrieved on 29 September 2026. Web caching and ongoing repository edits can make displayed snapshots differ; consequently no unverified commit-to-render parity is asserted. The exact labels above make the mathematical comparison reproducible. The inspection was targeted at these statements and their prerequisites, not an exhaustive audit of every file in the repository.

1.2 What is classical, and what is developed here

The background mechanisms are classical: Hahn summability, formal Lagrange inversion, filtered fixed points, and formal substitution. The standard transseries literature already develops composition and inversion in much larger settings [3, 5, 6]. Gessel gives a detailed account of Lagrange inversion [4]. More recent work treats Taylor expansions for generalized series and systematic correspondences between summable derivations and automorphisms [7, 8, 9]. None of these general subjects is claimed as a new discovery here.

The contribution is a precise resolution of the listed questions in the repository’s chosen coefficient algebra, with explicit support bounds, a necessary-and-sufficient fixed-alphabet criterion, and an all-sector leading coefficient theorem. The proofs also distinguish three notions that must not be conflated: existence of a formal coefficient, finiteness of an entire truncation, and analytic convergence after evaluation. No literature-priority claim is made for the individual formulas or for the classification in a possibly different existing notation.

We work over $\mathbb{R}$ to make all real exponents and the constants e^b unambiguous. The purely power–logarithmic results work over any characteristic-zero field containing the exponent group. Extending constants is not the issue addressed here.

2 A support calculus for Hahn power–log series

2.1 The coefficient ring and its two kinds of finiteness

Fix an additive subgroup $\Gamma \subseteq \mathbb{R}$ containing 1. Write

$$X = t^{-1}, \quad L = \log X$$

as formal notation. Define $\mathcal{C}_\Gamma$ to consist of all expressions

$$f = \sum_{\gamma \in S} p_\gamma(L) t^\gamma, \quad p_\gamma \in \mathbb{R}[L], \quad (1)$$

where $S \subseteq \Gamma$ is well ordered in the usual order and zero coefficients are omitted. The notation $\mathbb{R}[L]((t^\Gamma))$ will mean this *Hahn ring*, not a ring restricted to a discrete lattice. Each block is a polynomial in L , but its degree can grow without a uniform bound. For $f \neq 0$ set

$$v(f) = \min \text{supp } f, \quad \text{lc}_t(f) = p_{v(f)}(L), \quad v(0) = +\infty.$$

The subring $\mathcal{C}_{\Gamma, \text{lf}}$ consists of the *left-finite* elements: for every $B \in \mathbb{R}$, only finitely many support exponents are at most B .

A family $(f_i)_{i \in I}$ is *Hahn summable* if its union of supports is well ordered and, for every γ , only finitely many i have a nonzero t^γ block. Its sum is then defined blockwise. This definition does not sum infinitely many real contributions to a single coefficient, even when those contributions converge numerically.

Lemma 2.1 (Support lemma). *If $S, T \subseteq \mathbb{R}$ are well ordered, then $S + T$ is well ordered and every real number has only finitely many representations $s + t$ with $(s, t) \in S \times T$. If $S \subseteq (0, \infty)$ is nonempty and well ordered, and $\delta = \min S$, its additive monoid*

$$M(S) = \{0\} \cup \bigcup_{r \geq 1} \underbrace{(S + \cdots + S)}_{r \text{ times}}$$

is well ordered. Every target has only finitely many representations as an ordered word in S . If S is left finite, then $M(S)$ is left finite.

Proof. An infinite family of different representations of one target would contain an increasing sequence of first summands and hence a decreasing sequence of second summands, impossible in a well-ordered set. If $S + T$ had a strictly decreasing sequence, choose one representation of each term. A sequence in a well-ordered set has an infinite nondecreasing subsequence; along such a subsequence of first summands, the second summands would strictly decrease. This proves the first assertion and, by induction, its finite-sum analogue.

A word of length r in positive elements of S has sum at least $r\delta$. Below any fixed upper bound, only finitely many lengths can therefore occur. For these finitely many lengths the preceding assertions apply. A nonempty subset of $M(S)$ has a least element by first restricting below any one of its elements and taking a minimum in a finite union of well-ordered sets. The same length bound gives finiteness of the ordered words for a prescribed sum. In the left-finite case, a word with sum at most B uses only the finitely many letters of $S \cap (0, B]$, and has bounded length. There are only finitely many such words. $\square$

Lemma 2.2 (Elementary algebra and summation). *The Cauchy product makes $\mathcal{C}_\Gamma$ a commutative integral domain, and $v(fg) = v(f) + v(g)$. An element $f \neq 0$ is a unit if and only if $\text{lc}_t(f)$ is a nonzero real constant. If $v(f_j) \rightarrow +\infty$, the sequence-family (f_j) is Hahn summable. Every sequence whose successive errors have valuation tending to $+\infty$ converges in the valuation topology. These statements restrict to $\mathcal{C}_{\Gamma, \text{lf}}$ when all the supports involved are left finite.*

Proof. The support lemma makes each product coefficient finite. The leading block of a product is the product of the leading blocks, proving the valuation identity. An inverse forces those leading blocks to be units of $\mathbb{R}[L]$, which are precisely the nonzero constants. Conversely, factor a candidate unit as $ct^\gamma(1+u)$ with $v(u) > 0$ and use $\sum_{j \geq 0} (-u)^j$, whose summability follows from the support lemma.

If the valuations of a family indexed by $j \in \mathbb{N}$ tend to infinity, below any fixed bound only finitely many of its supports occur. Finite unions are well ordered, and each coefficient has a finite fiber. This also proves the usual Cauchy-sequence assertion by stabilized initial segments, or by a subsequence with successively increasing error valuations. Left finiteness is preserved because at a fixed bound there are only finitely many input supports, each with finitely many relevant exponents. $\square$

2.2 Differentiation and near-identity substitution

Define a strong derivation by

$$D_0(t^\gamma p(L)) = t^{\gamma+1}(p'(L) - \gamma p(L)). \quad (2)$$

It obeys $D_0 X = 1$, $D_0 L = t$, and

$$v(D_0^j f) \geq v(f) + j. \quad (3)$$

Here and below an inequality involving $v(0)$ has the usual extended-value meaning. Repeated application gives the block identity

$$D_0^j(t^\beta p(L)) = t^{\beta+j} \prod_{i=0}^{j-1} (\partial_L - \beta - i) p(L). \quad (4)$$

An empty product of operators is the identity.

For $B \in \mathcal{C}_\Gamma$ with $v(B) > -1$, put $u = tB$. Thus $v(u) > 0$. Define substitution by $H = X + B$ on the generators by

$$t^\gamma \circ H = t^\gamma(1+u)^{-\gamma}, \quad L \circ H = L + \log(1+u). \quad (5)$$

Binomial and logarithmic series in u are Hahn summable. For a general $f \in \mathcal{C}_\Gamma$ this gives the useful equivalent formula

$$f \circ (X + B) = \sum_{j \geq 0} \frac{B^j}{j!} D_0^j f. \quad (6)$$

The summand has valuation at least $v(f) + j(1+v(B))$, so the outer sum is summable, including for supports with infinitely many elements in a bounded interval. For each fixed j , the support lemma justifies its products and its termwise differentiation.

Proposition 2.3 (Substitution estimates). *The rule (6) is a strongly summation-preserving algebra homomorphism. It preserves the leading block and the valuation of nonzero f . It satisfies the chain rule and associative composition. If $v(B), v(C) \geq \alpha > -1$, then*

$$v(f \circ (X + B) - f \circ (X + C)) \geq v(B - C) + v(f) + 1. \quad (7)$$

The maps $X + B$ with $v(B) > -1$ form a group under composition.

Proof. The $j \geq 1$ terms in (6) have strictly larger valuation than f , proving preservation of its leading block. The Leibniz rule and the binomial identity prove multiplicativity after collecting summable terms. The identities in (5), and the identities for logarithms and powers of $1+u$, prove associativity on each monomial and logarithmic block. The same identities differentiated

prove the chain rule. To pass to Hahn sums, note that the union of input supports has a common lower bound, so the Taylor index is bounded for each target exponent; the support lemma then proves strong summability and permits the rearrangements.

For the difference estimate use $B^j - C^j = (B - C) \sum_{i=0}^{j-1} B^i C^{j-1-i}$. Its contribution at index $j \geq 1$ has valuation at least

$$v(B - C) + v(f) + 1 + (j - 1)(\alpha + 1),$$

which proves (7). Inverting $X + A$ amounts to solving $B = -A \circ (X + B)$. This map preserves $v(B) \geq v(A) = \alpha$ and improves difference valuation by at least $\alpha + 1 > 0$. Iteration from 0 therefore converges by Lemma 2.2; the same bound proves uniqueness. Associativity gives two-sided inversion: constructing a right inverse for the right inverse and associating the two identities shows that the first right inverse is also a left inverse. Closure under composition follows directly from preservation of valuation and the formula $(X + A) \circ (X + B) = X + B + A \circ (X + B)$. $\square$

Remark 2.4 (The Archimedean hypothesis has a job). We use $\Gamma \subseteq \mathbb{R}$, not an arbitrary ordered abelian group. For every $\delta > 0$, the multiples $j\delta$ are cofinal in $\mathbb{R}$. That fact justifies the bounds on the Taylor and iteration indices. Replacing Γ by a higher-rank ordered group without changing the argument would create a gap.

3 Dense exponents: the ordinary Lagrange sum is enough

3.1 A formal coefficient lemma

We record the algebraic form of classical Lagrange inversion to isolate exactly where the index set comes from [4].

Lemma 3.1 (Auxiliary-parameter inversion). *Let R be a commutative $\mathbb{Q}$ -algebra, $\phi(z), \psi(z) \in R[[z]]$, and let $U(s) \in sR[[s]]$ be the unique solution of $U = s\phi(U)$. For $n \geq 1$,*

$$[s^n]\psi(U) = \frac{1}{n}[z^{n-1}]\psi'(z)\phi(z)^n. \quad (8)$$

No invertibility hypothesis on $\phi(0)$ is necessary.

Proof. Existence and uniqueness follow coefficient by coefficient. At a fixed order both sides of (8) are universal polynomials, over $\mathbb{Q}$, in finitely many coefficients of ϕ and ψ . It is therefore enough to work in the universal polynomial ring with $\phi(0)$ inverted, where $s = z/\phi(z)$ is an invertible change of local parameter. The formal residue change of variables gives

$$\begin{aligned} [s^n]\psi(U) &= \operatorname{Res}_{z=0} \psi(z) \left(\frac{z}{\phi(z)} \right)^{-n-1} \left(\frac{z}{\phi(z)} \right)' dz \\ &= \operatorname{Res}_{z=0} \psi(z) \left(z^{-n-1} \phi^n - z^{-n} \phi^{n-1} \phi' \right) dz \\ &= \frac{1}{n} \operatorname{Res}_{z=0} z^{-n} \psi'(z) \phi(z)^n dz. \end{aligned}$$

The last step uses the zero residue of the derivative of $\psi(z)\phi(z)^n z^{-n}$. The universal polynomial ring injects into its localization, so the polynomial identity already holds before inversion. Specialization proves it for the original R . $\square$

3.2 The summable formula and its coefficient bound

Theorem 3.2 (Hahn Lagrange–Bürmann reversion). *Let $A \neq 0$ belong to $\mathcal{C}_\Gamma$, put $\alpha = v(A) > -1$ and $\delta = \alpha + 1$, and let $F = X + A$. Then*

$$F^{(-1)} = X + B, \quad B = \sum_{r \geq 1} \frac{(-1)^r}{r!} D_0^{r-1}(A^r). \quad (9)$$

This is a Hahn-summable, positive-integer-indexed family. Its r th term has valuation at least $r\delta - 1$. Consequently the coefficient of t^λ receives contributions only from

$$1 \leq r \leq \left\lfloor \frac{\lambda + 1}{\delta} \right\rfloor. \quad (10)$$

Writing $A = \sum p_\gamma t^\gamma$, that coefficient is explicitly

$$[t^\lambda]B = \sum_{1 \leq r \leq \lfloor (\lambda+1)/\delta \rfloor} \frac{(-1)^r}{r!} \left\{ \prod_{j=0}^{r-2} (\partial_L - \lambda + r - 1 - j) \right\} [t^{\lambda-r+1}]A^r. \quad (11)$$

The formula remains valid when the support of A is not left finite.

Proof. Apply Lemma 3.1 in $R = \mathcal{C}_\Gamma$, using the formal Taylor series

$$\phi(z) = - \sum_{j \geq 0} \frac{D_0^j A}{j!} z^j, \quad \psi(z) = z.$$

The Taylor map to $R[[z]]$ is an algebra homomorphism. Thus

$$[s^r]U(s) = \frac{(-1)^r}{r!} D_0^{r-1}(A^r).$$

This first proves the identity with the auxiliary parameter s ; it makes no claim about evaluating an arbitrary formal power series at $s = 1$.

For these particular coefficients, (3) gives

$$v(D_0^{r-1}(A^r)) \geq r\alpha + r - 1 = r\delta - 1 \longrightarrow +\infty.$$

Therefore $U(1)$ is a legitimate Hahn sum. The identity $U(s) = -sA \circ (X + U(s))$ may be specialized to $s = 1$: in a term with j factors of U , the normalized valuation is bounded below by δ times the sum of their positive auxiliary indices. Only finitely many such total indices occur at a fixed target exponent, and every remaining multiplication has finite fibers by Lemma 2.1. Thus all required rearrangements are summable. The specialized equation is precisely the inverse equation. Proposition 2.3 proves uniqueness and two-sided inversion. Formula (11) follows from (4), with $\beta = \lambda - r + 1$. $\square$

Corollary 3.3 (A support certificate). *Let $S = \text{supp}(A) + 1 \subseteq (0, \infty)$ and $M = M(S)$. Then*

$$\text{supp}(B) + 1 \subseteq M \setminus \{0\}. \quad (12)$$

If $A \in \mathcal{C}_{\Gamma, \text{lf}}$, then $B \in \mathcal{C}_{\Gamma, \text{lf}}$. To determine B below an exclusive cutoff T , it is enough to keep indices $r < (T + 1)/\delta$ in (9).

Proof. The support of the r th term of (9) is contained in $r \text{supp}(A) + (r - 1) = rS - 1$. Differentiation can delete a block but cannot create an exponent other than the displayed shift. Lemma 2.1 now proves the assertions. $\square$

Remark 3.4 (What the repository's dense-exponent question conflates). The outer summation index in Lagrange inversion counts applications of a perturbation; it does not enumerate the exponent group. Under the repository's $v(A) \geq 0$ hypothesis, $r \leq \lambda + 1$ remains a valid bound for a coefficient at any real exponent λ . Density does not force a transfinite or noninteger perturbation index. What may fail is the *finiteness of all the coefficients below a cutoff*: this fails for non-left-finite supports, not for the integer-indexed operator formula.

3.3 Examples separating the issues

Example 3.5 (Irrational exponents). Let $\rho > 0$ be irrational and $A = t^\rho$, in $\Gamma = \mathbb{Z} + \rho\mathbb{Z}$. The group is dense, but reversion starts

$$F^{(-1)} = X - t^\rho - \rho t^{2\rho+1} - \frac{\rho(3\rho+1)}{2} t^{3\rho+2} - \frac{\rho(4\rho+1)(2\rho+1)}{3} t^{4\rho+3} + \dots$$

Every exponent lies in the single ray $r(\rho+1) - 1$. A dense ambient group need not produce a complicated support in an individual computation.

Example 3.6 (An infinite bounded initial segment). In $\Gamma = \mathbb{Q}$, take

$$A = \sum_{n \geq 2} t^{1-1/n}.$$

Its support is increasing and well ordered, with minimum $1/2$ and limit 1 . It is not left finite. Theorem 3.2 still applies, but the truncation of B below 1 is $-A$ and contains infinitely many blocks. Indeed every $r \geq 2$ term has valuation at least 2 . Finite outer index range is not a finite-data algorithm for an arbitrary Hahn input.

Example 3.7 (A sublinear but unbounded perturbation). For $A = t^{-1/2} = \sqrt{X}$, the theorem applies with $\delta = 1/2$. The inverse is the familiar positive algebraic branch

$$F^{(-1)} = X + \frac{1}{2} - \frac{1}{2}\sqrt{1+4X} = X - X^{1/2} + \frac{1}{2} - \frac{1}{8}X^{-1/2} + \frac{1}{128}X^{-3/2} - \dots$$

Here a fixed coefficient can require more Lagrange indices than the integer-scale bound $r \leq N+1$. The correct quantity is $v(tA) = \delta$.

4 Newton reversion preserves the support monoid

Let $F = X + A$ satisfy Theorem 3.2, and put $H_* = F^{(-1)}$. Define

$$H_0 = X, \quad H_{k+1} = H_k - \frac{F \circ H_k - X}{F' \circ H_k}, \quad F' = D_0 F. \quad (13)$$

Every denominator is a unit with leading block 1 because $v(A') \geq \alpha + 1 = \delta > 0$.

Theorem 4.1 (No exponent-lattice enlargement). *With $S = \text{supp}(A) + 1$ and $M = M(S)$, every iterate of (13) is well defined in $\mathcal{C}_\Gamma$ and satisfies*

$$\text{supp}(t(H_k - X)) \subseteq M \setminus \{0\}. \quad (14)$$

If $e_k = v(H_k - H_)$ is finite, then*

$$e_{k+1} \geq 2e_k + \alpha + 2, \quad e_k \geq (2^{k+1} - 1)\delta - 1. \quad (15)$$

The iterates converge to H_ in the valuation topology.*

Proof. First prove the support assertion, which is stronger than membership in Γ . Write $H = X + t^{-1}V$, where $\text{supp } V \subseteq M \setminus \{0\}$. For one input block $t^\gamma p_\gamma(L)$,

$$t(t^\gamma p_\gamma(L) \circ H) = t^{\gamma+1}(1+V)^{-\gamma} p_\gamma(L + \log(1+V)).$$

The support on the right is contained in $(\gamma+1) + M$. Thus $tA \circ H$ and $t(F \circ H - X) = V + tA \circ H$ have support in $M \setminus \{0\}$. Similarly $A' \circ H$ has support there. The inverse of $1 + A' \circ H$ is a geometric series supported in M , so the normalized Newton correction has support in $M \setminus \{0\}$. Induction starts with $V = 0$. Since $\min(M \setminus \{0\}) = \delta$, all iterates satisfy $v(H_k - X) \geq \alpha$.

For the rate, put $E = H - H_*$ and $e = v(E) \geq \alpha$. Taylor-expand F at H with displacement $-E$:

$$0 = F \circ H - X - E(F' \circ H) + \sum_{j \geq 2} \frac{(-E)^j}{j!} (F^{(j)} \circ H).$$

For $j \geq 2$, $F^{(j)} = D_0^j A$ has valuation at least $\alpha + j$; composition preserves it. The last sum has valuation at least $\alpha + 2e + 2$, since $e + 1 \geq \delta > 0$. Dividing by the slope unit gives the first inequality in (15). Also $v(H_0 - H_*) = v(B) = \alpha$, because the first Lagrange term is $-A$ and all others have strictly higher valuation. Writing $p_k = e_k + 1$ gives $p_{k+1} \geq 2p_k + \delta$ and $p_0 = \delta$, proving the second bound. An exact iterate remains exact; otherwise these bounds tend to infinity. $\square$

Corollary 4.2 (The Puiseux denominator question). *For $s \in \mathbb{N}_{>0}$ and $A \in \mathbb{R}[L][[t^{1/s}]]$, every Newton iterate and the inverse belong to $\mathbb{R}[L]((t^{1/s}))$. Thus the repository question has answer $\sigma(s) = s$. Any integer-valued universal choice of $\sigma(s)$ must be divisible by s .*

Proof. Here $S \subseteq s^{-1}\mathbb{Z}$ and $1 \in s^{-1}\mathbb{Z}$, so (14) and (12) prove the inclusion. For minimality, take $A = t^{1/s}$. Its inverse starts with $X - t^{1/s}$, and even the first Newton iterate has that correction as its leading block. The exponent $1/s$ belongs to $\sigma^{-1}\mathbb{Z}$ only if s divides σ . $\square$

Remark 4.3 (Why incompleteness of the Puiseux union is irrelevant here). A Cauchy sequence with input denominators tending to infinity may leave the Puiseux union, as the repository correctly explains. But a Newton orbit from one fixed input is not an arbitrary sequence in that union. Its support is trapped in the invariant monoid above. The division step is by a unit whose geometric inverse cannot introduce a new exponent denominator.

5 A complete one-exponential sector algebra

5.1 Definition, summation, and topology

Fix $\mu > 0$ and introduce the formal symbol q , with the intended interpretation $q = e^{-\mu X}$. Set

$$\mathcal{E}_{\Gamma, \mu} = \mathcal{C}_\Gamma[[q]], \quad D = D_0 - \mu q \frac{\partial}{\partial q}. \quad (16)$$

Thus

$$D \left(\sum_{n \geq 0} q^n f_n \right) = \sum_{n \geq 0} q^n (D_0 - n\mu) f_n. \quad (17)$$

There is no uniform lower bound on the valuations $v(f_n)$ as n varies. For instance, $\sum_{n \geq 0} q^n t^{-n^2}$ is an element of $\mathcal{E}_{\Gamma, \mu}$.

A family in $\mathcal{E}_{\Gamma, \mu}$ is called *sectorwise Hahn summable* when its coefficient family at each q^n is Hahn summable in $\mathcal{C}_\Gamma$. Give $\mathcal{E}_{\Gamma, \mu}$ the product topology of the valuation topologies of its coefficient copies of $\mathcal{C}_\Gamma$. A neighborhood can impose finitely many conditions $v(f_n) > B_n$.

This topology lets both increasingly small power-log terms in a fixed sector and tails with increasingly large sector index tend to zero. It must not be identified with the q -adic topology with discrete coefficients, or with a lexicographic valuation topology. We use q -adic order for exact sector estimates as well; q -adic convergence implies convergence in the product topology.

The product topology alone does not describe every Hahn sum in a non-left-finite coefficient ring. Sectorwise Hahn summability remains the formal summation rule. Keeping these two structures explicit avoids a hidden analytic convergence assumption.

Proposition 5.1 (Arithmetic and completeness). *The ring $\mathcal{E}_{\Gamma,\mu}$ is a complete topological differential algebra for the coefficientwise valuation topology. Its multiplication and derivation respect sectorwise Hahn summation. An element $f = \sum q^n f_n$ is a unit precisely when f_0 is a unit of $\mathcal{C}_\Gamma$, equivalently when $f_0 \neq 0$ and $\text{lc}_t(f_0) \in \mathbb{R}^\times$. Exponentials and logarithms of q -adically small elements are defined by their usual power series. More generally, $\exp(g)$ is well defined if $v(g_0) > 0$, by separating g_0 and the positive-sector tail.*

Proof. A coefficient of a product is a finite sum over sector pairs $i + j = n$. The support lemma applies within each pair. Completeness is coordinatewise completeness of $\mathcal{C}_\Gamma$. Formula (17) proves continuity and compatibility with summation. A unit must have invertible constant q -coefficient; conversely factor $f = f_0(1 + h)$ with $h \in q\mathcal{E}_{\Gamma,\mu}$ and use the q -adically summable geometric series. The exponential statements follow from $\text{ord}_q(h^r) \geq r$ for $h \in q\mathcal{E}_{\Gamma,\mu}$, and from the positive valuation of g_0 . For general g with $v(g_0) > 0$, define $\exp(g) = \exp(g_0) \exp(g - g_0)$. $\square$

Remark 5.2 (Why one fixed scalar weight does not suffice). Suppose a proposed weight assigns $q^n t^\gamma$ the number $an + b\gamma$ with $b > 0$. The support of $\sum q^n t^{-n^2}$ has weights $an - bn^2$ unbounded below. Thus no fixed scalar weight of this kind describes all of $\mathcal{E}_{\Gamma,\mu}$ as a bounded-below series algebra. The product/sectorwise construction does not impose an artificial affine growth restriction on negative power orders. A smaller, weight-bounded algebra can be useful for algorithms, but is a different object.

5.2 Antiderivatives do not require further exponential scales

Although $Dq = -\mu q$ does not improve the power order, the positive-sector differential operators are particularly easy to invert.

Theorem 5.3 (Sectorwise integration). *The derivation $D : \mathcal{E}_{\Gamma,\mu} \rightarrow \mathcal{E}_{\Gamma,\mu}$ is surjective and has kernel $\mathbb{R}$. For every $n \geq 1$, the operator $D_0 - n\mu$ on $\mathcal{C}_\Gamma$ is bijective, with*

$$(D_0 - n\mu)^{-1}f = - \sum_{j \geq 0} \frac{D_0^j f}{(n\mu)^{j+1}}. \quad (18)$$

The series is Hahn summable and restricts to the left-finite coefficient class. The zero-sector antiderivative is the usual power-log block antiderivative, unique up to a real constant.

Proof. The summands in (18) have valuations at least $v(f) + j$, so the series is summable. Multiplication by $D_0 - n\mu$ telescopes, leaving f . The operator is injective because, for $g \neq 0$, the term $-n\mu g$ has valuation $v(g)$ whereas $D_0 g$ has strictly higher valuation.

For the zero sector, integrate a block $t^\gamma p(L)$ by seeking $t^{\gamma-1}Q(L)$. The equation is

$$Q' - (\gamma - 1)Q = p.$$

If $\gamma \neq 1$, its unique polynomial solution is

$$Q = - \sum_{j=0}^{\deg p} \frac{p^{(j)}}{(\gamma - 1)^{j+1}}.$$

At $\gamma = 1$, take the polynomial primitive of p , with zero constant. The supports are shifted by -1 , hence remain well ordered. The only kernel element occurs at exponent 0 and is a real constant. Applying this argument independently in every sector proves the theorem. $\square$

6 The exact fixed-alphabet substitution criterion

Consider a prospective change of variable

$$G = X + P(L) + U + h, \quad P \in \mathbb{R}[L], \quad v(U) > 0, \quad h \in q\mathcal{E}_{\Gamma, \mu}. \quad (19)$$

We allow $U = 0$ with $v(0) = +\infty$. Substitution of G into any base element of $\mathcal{C}_\Gamma$ is already meaningful: first substitute $g_0 = X + P(L) + U$, then Taylor-expand in h . At a fixed output sector only finitely many Taylor powers of h occur.

What remains is the image of q . The interpretation $q = e^{-\mu X}$ requires it to obey

$$D(q \circ G) = -\mu(DG)(q \circ G). \quad (20)$$

We require a nonzero image. This rules out the algebra homomorphism that simply kills every positive sector, which is not a change of variable.

Theorem 6.1 (Exact one-exponential admissibility). *For G as in (19), there is a strongly sectorwise-summation-preserving substitution homomorphism on $\mathcal{E}_{\Gamma, \mu}$ extending the base substitution, with nonzero image of q and the chain rule*

$$D(f \circ G) = (DG)((Df) \circ G),$$

if and only if

$$P(L) = aL + b \quad \text{and} \quad m := \mu a \in \Gamma. \quad (21)$$

When (21) holds, the normalized substitution is

$$q \circ G = e^{-\mu b} q t^m \exp(-\mu(U + h)). \quad (22)$$

The normalization means that the leading block of its q -coefficient is $e^{-\mu b} t^m$. It makes the substitution unique. Without normalization, the image of q can be multiplied by an arbitrary nonzero real constant.

Proof of necessity. Let $W = q \circ G \neq 0$, and let $k = \text{ord}_q W \geq 0$. Write $W = q^k u + \mathcal{O}(q^{k+1})$, $u \in \mathcal{C}_\Gamma \setminus \{0\}$. The leading sector of (20) gives

$$D_0 u = \mu(k-1)u - \mu(tP'(L) + D_0 U)u. \quad (23)$$

If $k \neq 1$, the right side has valuation $v(u)$ whereas the left side has valuation at least $v(u) + 1$, a contradiction. Thus $k = 1$. Set $V = u \exp(\mu U)$. Its defining exponential is available because $v(U) > 0$, and (23) becomes

$$D_0 V = -\mu t P'(L) V.$$

Write its leading block as $t^\gamma p(L)$ with $\gamma \in \Gamma$ and $p \neq 0$. Comparing the block at exponent $\gamma + 1$ yields

$$p' = (\gamma - \mu P')p. \quad (24)$$

If $\gamma - \mu P'$ were a nonzero polynomial, its product with p would have degree at least $\deg p$, while p' has degree strictly smaller or vanishes. Hence $\gamma - \mu P' = 0$ and $p' = 0$. Therefore $P = aL + b$ and $\gamma = \mu a \in \Gamma$. The same differential equation and the kernel calculation for D_0 imply $V = ct^\gamma$ for a nonzero real c . $\square$

Proof of sufficiency and uniqueness. Assume (21). The right side of (22) is a well-defined element of $q\mathcal{E}_{\Gamma,\mu}$ with an invertible leading sector coefficient. Extend base substitution by

$$\left(\sum_{n \geq 0} q^n f_n\right) \circ G = \sum_{n \geq 0} (q \circ G)^n (f_n \circ G). \quad (25)$$

In sector N only $n \leq N$ can occur; for each n , Taylor expansion of f_n around g_0 uses only finitely many powers of h . Within each such term Proposition 2.3 preserves Hahn summability. This proves well-definedness, multiplicativity, and strong compatibility with sectorwise Hahn sums. Direct differentiation of (22) proves (20); the base chain rule then extends to every element.

For uniqueness, compare any other nonzero solution W of (20) with the canonical one W_0 . Necessity proved that both have sector order 1 and that the coefficient of q is a unit of $\mathcal{C}_\Gamma$. Thus W/W_0 belongs to $\mathcal{E}_{\Gamma,\mu}$ (divide after canceling q), and its derivative is zero. Theorem 5.3 implies that this quotient is a real constant. Normalization makes that constant 1. $\square$

Corollary 6.2 (A genuine obstruction, not a convergence defect). *For the ordinary integer-power algebra $\mathbb{R}[L]((t))[[q]]$, the map $X + a \log X + b + U + h$ is admissible precisely when $\mu a \in \mathbb{Z}$. The map $X + L^2$ is not admissible for any additive exponent subgroup $\Gamma \subseteq \mathbb{R}$. Neither changing the topology on the same algebra nor allowing all real power exponents removes the latter obstruction.*

Proof. The first statement specializes Theorem 6.1. For $P = L^2$, (24) is impossible for every γ . This is a polynomial differential equation obstruction inside the coefficient ring, not a failure to choose sufficiently fine truncations. Formally the missing factor is $e^{-\mu L^2}$, which is not a power-logarithmic block series of the prescribed kind. $\square$

Corollary 6.3 (Minimal power-group enlargement). *To permit a specified finite set of affine logarithmic shifts with coefficients $a_1, \dots, a_r$, the smallest additive power group containing the original Γ is*

$$\Gamma' = \Gamma + \mathbb{Z}\mu a_1 + \dots + \mathbb{Z}\mu a_r.$$

The same assertion holds for any specified set of shifts by taking its additive span. No enlargement of the power group alone permits a polynomial logarithmic shift of degree at least two.

Proof. Necessity follows by applying the criterion separately to each shift. Sufficiency follows by constructing the algebra over Γ' and applying the theorem. The final assertion was just proved. $\square$

Corollary 6.4 (Maximality among sublinear base displacements). *More generally, let $G = X + A + h$ with $A \in \mathcal{C}_\Gamma$, $v(A) > -1$, and $h \in q\mathcal{E}_{\Gamma,\mu}$. It admits a substitution of the kind in Theorem 6.1 if and only if*

$$A = aL + b + U, \quad \mu a \in \Gamma, \quad v(U) > 0.$$

In particular, a power displacement such as $X + \sqrt{X}$ cannot be admitted in this fixed one-exponential algebra, even when $1/2 \in \Gamma$.

Proof. Repeat the leading-sector argument with $D_0 A$ in place of $tP' + D_0 U$. Since $v(D_0 A) > 0$, the same comparison first forces $\text{ord}_q(q \circ G) = 1$. Its leading coefficient $u \neq 0$ then satisfies $D_0 u = -\mu(D_0 A)u$. If $\alpha = v(A) < 0$, the leading block of $D_0 A$ cannot vanish: its polynomial coefficient is $p' - \alpha p$ for a nonzero polynomial p , and $\alpha \neq 0$. Thus

$$v((D_0 A)u) = v(u) + \alpha + 1 < v(u) + 1 \leq v(D_0 u),$$

a contradiction. Hence $v(A) \geq 0$, and $A = P(L) + U$ with $v(U) > 0$. Theorem 6.1 finishes both directions. The case $A = 0$ is included by taking $a = b = U = 0$. $\square$

7 A substitution group and a core-normalized inverse

7.1 The group structure

Let $\mathcal{G}_{\Gamma,\mu}$ be the set of maps

$$G = X + aL + b + U + h, \quad \mu a \in \Gamma, \quad v(U) > 0, \quad h \in q\mathcal{E}_{\Gamma,\mu}, \quad (26)$$

with the normalized substitution of Theorem 6.1. Let $\mathcal{G}_0$ be the same class with $h = 0$, and $\mathcal{G}_q = \{X + h : h \in q\mathcal{E}_{\Gamma,\mu}\}$.

Theorem 7.1 (Closed substitution group). *The class $\mathcal{G}_{\Gamma,\mu}$ is a group under composition. Its substitutions are continuous automorphisms of $\mathcal{E}_{\Gamma,\mu}$ for the product topology. Reduction to sector zero gives a split exact sequence*

$$1 \longrightarrow \mathcal{G}_q \longrightarrow \mathcal{G}_{\Gamma,\mu} \longrightarrow \mathcal{G}_0 \longrightarrow 1. \quad (27)$$

In particular, this is a semidirect product of the positive-sector group and the admissible base group. The pair (a, b) is additive under composition and changes sign under inversion.

Proof. First, the base near-identity group exists by Proposition 2.3. Composing two base maps adds their zero-power polynomials $aL + b$; all changes to L are of positive power valuation. Thus $\mathcal{G}_0$ is closed under composition and its inverses have the form $X - aL - b + V$, $v(V) > 0$. The lattice condition is preserved.

For $h \in q\mathcal{E}_{\Gamma,\mu}$, full substitution by $X + h$ is simply

$$f \circ (X + h) = \sum_{j \geq 0} \frac{h^j}{j!} D^j f. \quad (28)$$

The derivation preserves sector order, so the j th term has sector order at least $\text{ord}_q f + j$. Taylor identities prove associativity and the chain rule with exact finite algebra in each output sector. The equation for the inverse correction $b = -h \circ (X + b)$ is a q -adic contraction: its difference at two inputs has sector order at least one greater than the difference of the inputs. Its unique solution gives a two-sided inverse as in Proposition 2.3. Hence $\mathcal{G}_q$ is a group.

Every map $G = g_0 + h$ in (26) factors uniquely as

$$G = (X + k) \circ g_0, \quad k = h \circ g_0^{\langle -1 \rangle} \in q\mathcal{E}_{\Gamma,\mu}.$$

The right-hand side has the canonical substitution of Theorem 6.1: on base elements this is ordinary Taylor substitution, and on q the exponential factors multiply to (22). Equivalently, the chain rule and the normalized leading q -block determine that image uniquely. This verifies associativity across the two classes. Sector-zero reduction is a homomorphism with kernel $\mathcal{G}_q$ and with splitting the inclusion of $\mathcal{G}_0$. Conjugating a positive-sector map by a base map again reduces to X in sector zero, so the kernel is normal. This proves (27) and inversion of every admissible map.

For continuity, fix one output sector. Only finitely many input sectors enter (25); in each of these, only finitely many powers of the positive-sector displacement enter. Base substitution preserves valuation, differentiation has a lower valuation bound, and multiplication by a fixed coefficient shifts valuation by a fixed amount. A sufficiently high input cutoff therefore implies any prescribed output cutoff. The same argument applies to the inverse substitution. $\square$

7.2 Why the unnormalized Lagrange formula can fail

Example 7.2 (A constant shift creates infinitely many identical monomials). Take $F = X + b + q$ with $b \neq 0$. In the naive expression

$$\sum_{r \geq 1} \frac{(-1)^r}{r!} D^{r-1}(b + q)^r,$$

the coefficient of q in the r th summand is

$$-\frac{(\mu b)^{r-1}}{(r-1)!}.$$

Thus the same monomial q occurs at every r . This is not a sectorwise Hahn-summable family. Its real numerical sum happens to converge, but that is a different summation operation. The underlying map is nonetheless admissible and invertible. The obstruction is to using the original power-order proof unchanged, not to the existence of the inverse.

The cure is to invert the entire zero sector before expanding in the exponential tail. This is also the principle of the repository's exact-core inversion architecture; what must be supplied in the present algebra is summability and closure of every substitution.

Theorem 7.3 (Core-normalized Lagrange–Bürmann formula). *Let*

$$F = F_0 + E_+, \quad F_0 \in \mathcal{G}_0, \quad E_+ \in q\mathcal{E}_{\Gamma, \mu}, \quad H_0 = F_0^{(-1)}, \quad K = E_+ \circ H_0.$$

Then $K \in q\mathcal{E}_{\Gamma, \mu}$ and

$$F^{(-1)} = H_0 + \sum_{r \geq 1} \frac{(-1)^r}{r!} D^{r-1}(K^r D_0 H_0). \quad (29)$$

The family is q -adically summable. At sector q^N , only $r \leq N$ can contribute. If $\text{ord}_q E_+ = d > 0$, the sharper bound is $r \leq \lfloor N/d \rfloor$.

Proof. Since H_0 is admissible, K exists and has the same positive sector order as E_+ . The identity $F \circ H_0 = X + K$ implies $F^{(-1)} = H_0 \circ (X + K)^{(-1)}$. Apply Lemma 3.1 with the Taylor map for D , with $\phi(z) = -K(X + z)$ and $\psi(z) = H_0(X + z)$. The general coefficient formula gives

$$\frac{(-1)^r}{r!} D^{r-1}(K^r D H_0).$$

Here $DH_0 = D_0 H_0$. Because D preserves sector order, the displayed coefficient has sector order at least rd . Specialization of the auxiliary parameter at 1 is therefore valid by exact finiteness in each sector. This proves (29) without summing any infinite family into a single real coefficient. $\square$

Example 7.4 (Exact agreement with a Lambert inverse). For $F = X + b + q$, $H_0 = X - b$ and $K = e^{\mu b} q$. Formula (29) becomes

$$F^{(-1)} = X - b - \sum_{n \geq 1} \frac{n^{n-1}}{n!} \mu^{n-1} e^{\mu b n} q^n = X - b + \frac{1}{\mu} \mathcal{W}(-\mu e^{\mu b} q), \quad (30)$$

where $\mathcal{W}$ denotes the formal series inverse of $z \mapsto ze^z$ at 0. Substituting $y = X - b + u$ into $F(y) = X$ gives $\mu u e^{\mu u} = -\mu e^{\mu b} q$, independently checking the formula. Here each sector is obtained from exactly one summable Lagrange term, unlike the naive expansion in $b + q$.

8 The exact leading block of every inverse sector

The core-normalized formula yields more than existence. It gives a sharp support boundary for every positive sector of a single-exponential perturbation.

Theorem 8.1 (Cayley leading-block law). *Let*

$$F = F_0 + qf, \quad F_0 = X + aL + b + U \in \mathcal{G}_0, \quad f = t^\beta P(L) + \text{higher power exponents},$$

where $f \neq 0$, $\beta = v(f)$ and $P = \text{lc}_t(f) \neq 0$. Write $m = \mu a$ and $\kappa = \beta - m$. Then

$$F^{(-1)} = H_0 + \sum_{n \geq 1} q^n b_n, \quad H_0 = F_0^{(-1)},$$

has the following exact properties for every $n \geq 1$:

$$v(b_n) = n\kappa, \tag{31}$$

$$[t^{n\kappa}]b_n = -\frac{n^{n-1}}{n!} \mu^{n-1} e^{\mu b n} P(L)^n. \tag{32}$$

In particular the logarithmic degree of this leading block is $n \deg P$, and no positive sector vanishes.

Proof. The inverse base has the form $H_0 = X - aL - b + V$, $v(V) > 0$. Put

$$qf \circ H_0 = qk, \quad k = e^{\mu b} t^{-m} e^{-\mu V} (f \circ H_0).$$

Base substitution preserves leading blocks. Consequently

$$v(k) = \kappa, \quad \text{lc}_t(k) = e^{\mu b} P(L), \quad D_0 H_0 = 1 + \text{positive power order}.$$

Only $r = n$ contributes to q^n in (29). Using (17) gives the exact coefficient formula

$$b_n = \frac{(-1)^n}{n!} (D_0 - n\mu)^{n-1} (k^n D_0 H_0). \tag{33}$$

For any nonzero $g \in \mathcal{C}_\Gamma$, the leading block of $(D_0 - n\mu)g$ is $-n\mu$ times that of g , since $D_0 g$ has power order at least one higher. Apply this observation $n - 1$ times. The sign is $(-1)^n (-1)^{n-1} = -1$, giving (32) and proving its nonvanishing. The degree statement follows because $\mathbb{R}[L]$ is an integral domain. $\square$

8.1 Interpretation and a useful change of small parameter

The exponent pairs of the inverse satisfy $\gamma \geq n\kappa$ in sector n , and the line $\gamma = n\kappa$ is attained in every sector. This is an exact boundary, not a heuristic balance. A logarithmic shift in the base changes the slope from β to $\beta - \mu a$. When this number is negative, successively deeper exponential sectors have successively larger power prefactors; this is allowed in $\mathcal{E}_{\Gamma, \mu}$ and is precisely why a uniform lower power order was not imposed.

Introduce $Q = qt^\kappa$. Then all the coefficients of $F^{(-1)} - H_0 = \sum Q^n (t^{-n\kappa} b_n)$ have nonnegative power valuation. Their valuation-zero blocks have generating series

$$\sum_{n \geq 1} [t^{n\kappa}] b_n Q^n = \frac{1}{\mu} \mathcal{W}(-\mu e^{\mu b} P(L) Q). \tag{34}$$

This is an identity of formal leading-block series; it is not in general the full inverse. The tree coefficients n^{n-1} are classical [4]. Their precise occurrence here follows from the leading sector action $D_0 - n\mu \sim -n\mu$, while differentiation of power-log coefficients contributes only above the support boundary.

Remark 8.2 (What the theorem does and does not generalize). A tail with several primitive exponential sectors can have additional contributions and cancellations at a fixed sector. Formula (29) still applies, but the exact nonvanishing statement in Theorem 8.1 uses the hypothesis $E_+ = qf$. Likewise, an arbitrary formal coefficient f need not be realizable by an analytic function. Neither assumption may be silently removed.

9 A mixed logarithmic–exponential inverse with certified convergence

We now specialize to $\Gamma = \mathbb{Z}$, $\mu = 1$, and

$$F(X) = X + L + q. \quad (35)$$

This example exhibits all the structural features established above: the logarithmic core changes the leading power in each exponential sector, core inversion makes the coefficient formula finite, and the resulting formal series has an analytic realization with a quantitative error bound. The analytic proof is specific to this example and does not assert convergence of arbitrary elements of $\mathcal{E}_{\Gamma, \mu}$.

9.1 An exact rational coefficient formula

Let $w = H_0(X)$ be the formal inverse of $X + \log X$. Thus

$$w + \log w = X, \quad D_0 w = \frac{w}{w+1}, \quad q \circ w = qw. \quad (36)$$

Here $\log w = L + \log(tw)$ is legitimate because $tw = 1 +$ a positive-valuation series. The last identity follows from the normalized exponential substitution: $X - w = \log w$. Analytically, the positive solution is the Wright omega value $w = W(e^X)$; the repository already develops this core [1, 2].

Define the rational differential operator

$$T = \frac{w}{w+1} \frac{d}{dw}.$$

Theorem 7.3, with $K = qw$ and $D_0 H_0 = w/(w+1)$, gives the following exact formula:

$$F^{(-1)}(X) = w + \sum_{n \geq 1} q^n b_n(w), \quad b_n(w) = \frac{(-1)^n}{n!} (T - n)^{n-1} \frac{w^{n+1}}{w+1}. \quad (37)$$

Rational functions in this display are interpreted formally by their expansions at $w = \infty$ and then by substitution of $w = H_0(X)$. Their only possible finite poles will be at $w = -1$, so this interpretation is unambiguous. The first three coefficients are

$$b_1(w) = -\frac{w^2}{w+1}, \quad (38)$$

$$b_2(w) = -\frac{w^3(2w^2 + 2w - 1)}{2(w+1)^3}, \quad (39)$$

$$b_3(w) = -\frac{w^4(9w^4 + 18w^3 - 11w + 1)}{6(w+1)^5}. \quad (40)$$

The coefficient functions have no hidden infinite sum: obtaining b_n requires exactly $n - 1$ applications of a rational differential operator.

It is also useful to regard the exponential parameter as an independent complex variable z . If $\delta = \sum_{n \geq 1} b_n(w)z^n$, its defining identity is

$$\delta + \log \left(1 + \frac{\delta}{w} \right) + wze^{-\delta} = 0. \quad (41)$$

To see this, substitute $w + \delta$ into (35), use (36), and then replace q by z . The coefficient of δ at $(\delta, z) = (0, 0)$ is $1 + 1/w$, so the identity also determines all b_n recursively whenever $w \neq 0, -1$; for the analytic result below we use only real $w \geq 1$.

9.2 Sharp poles and exact growth at infinity

Theorem 9.1 (Two exact coefficient boundaries). *For every $n \geq 1$, b_n is rational, has no finite pole except $w = -1$, and has a pole of exact order $2n - 1$ there. With $(-1)!! = 1$,*

$$\lim_{w \rightarrow -1} (w + 1)^{2n-1} b_n(w) = -\frac{(2n-3)!!}{n!}. \quad (42)$$

At infinity,

$$b_n(w) = -\frac{n^{n-1}}{n!} w^n + \mathcal{O}(w^{n-1}). \quad (43)$$

Proof. The operator $T - n$ introduces no pole away from $w = -1$. Put $u = w + 1$. Then

$$T = \frac{-1 + u}{u} \frac{d}{du}.$$

Applied to a leading pole cu^{-k} , its most singular term is kcu^{-k-2} ; the other part of T , and multiplication by $-n$, produce strictly smaller pole orders. The initial function $w^{n+1}/(w+1)$ has leading pole $(-1)^{n+1}u^{-1}$. After $n-1$ applications, the pole order is $1 + 2(n-1)$ and its coefficient is

$$(-1)^{n+1}(1 \cdot 3 \cdot 5 \cdots (2n-3)).$$

Multiplication by $(-1)^n/n!$ proves (42), which is nonzero.

For the second assertion, the initial rational function is $w^n + \mathcal{O}(w^{n-1})$. The operator T lowers its leading power by one, whereas multiplication by $-n$ preserves it. Consequently the leading term after $n-1$ operations is $(-n)^{n-1}w^n$, and the prefactor in (37) gives (43). $\square$

Equation (43) is the specialization of Theorem 8.1 with $a = 1$, $\beta = 0$, $P = 1$, hence $\kappa = -1$. The inverse's n th exponential sector begins at t^{-n} . The exponential is still the asymptotically small quantity; its polynomial prefactor grows with n .

The deepest pole pieces alone have the formal sum

$$-\sum_{n \geq 1} \frac{(2n-3)!!}{n!} \frac{z^n}{(w+1)^{2n-1}} = (w+1) \left(\sqrt{1 - \frac{2z}{(w+1)^2}} - 1 \right). \quad (44)$$

This identity resums only the most singular Laurent term of each b_n , not the complete inverse. Analogous deepest-pole and square-root mechanisms already occur in the repository's special-function inversion chapter, notably `p6:thm:deepest-pole` [1]. The point here is an exact realization for the mixed logarithmic-exponential map and its compatibility with the support calculation above.

9.3 A uniform analytic sector bound

Theorem 9.2 (Convergence and explicit remainder). *For every real $w \geq 1$, equation (41) has a unique solution in $|\delta| \leq 1/4$ for $|z| \leq 1/(16w)$. This solution is holomorphic in z in the interior, has Taylor coefficients $b_n(w)$ from (37), and satisfies*

$$|b_n(w)| \leq \frac{1}{4}(16w)^n \quad (n \geq 1). \quad (45)$$

For $\theta = 16w|z| < 1$ and every integer $N \geq 0$,

$$\left| \delta(w, z) - \sum_{n=1}^N b_n(w) z^n \right| \leq \frac{\theta^{N+1}}{4(1-\theta)}. \quad (46)$$

In particular, set $x = w + \log w$ and $z = e^{-x}$. If $w \geq \log 32$, then $\theta = 16e^{-w} \leq 1/2$, the positive real inverse of $y \mapsto y + \log y + e^{-y}$ is

$$y = w + \sum_{n \geq 1} b_n(w) e^{-nx},$$

and its N -sector truncation has error at most

$$\frac{(16e^{-w})^{N+1}}{4(1-16e^{-w})} \leq \frac{1}{2}(16e^{-w})^{N+1}. \quad (47)$$

Proof. Use the principal logarithm on $|\delta/w| \leq 1/4$, put $s = 1 + 1/w$, and rewrite (41) as the fixed-point equation

$$\delta = \mathcal{T}_z(\delta) := -\frac{\log(1 + \delta/w) - \delta/w + wze^{-\delta}}{s}. \quad (48)$$

For $|u| < 1$,

$$|\log(1 + u) - u| \leq \frac{|u|^2}{2(1 - |u|)}.$$

Thus for $|\delta| \leq 1/4$, $w \geq 1$, and $|z| \leq 1/(16w)$,

$$|\log(1 + \delta/w) - \delta/w| \leq \frac{1}{24}, \quad |wze^{-\delta}| \leq \frac{e^{1/4}}{16} \leq \frac{1}{12}.$$

The last inequality uses $e^{1/4} \leq (1 - 1/4)^{-1} = 4/3$. Since $s \geq 1$, we obtain $|\mathcal{T}_z(\delta)| \leq 1/8$, strictly inside the proposed disc.

Its derivative with respect to δ satisfies

$$\begin{aligned} |\partial_\delta \mathcal{T}_z(\delta)| &\leq \frac{1}{s} \left(\left| \frac{1}{w + \delta} - \frac{1}{w} \right| + w|z|e^{1/4} \right) \\ &\leq \frac{1}{s} \left(\frac{1/4}{w(w - 1/4)} + \frac{1}{12} \right) \leq \frac{5}{12} < 1. \end{aligned}$$

The closed disc is convex, so this derivative estimate is a contraction bound. The contraction theorem supplies existence and uniqueness there. Picard iteration from zero consists of holomorphic functions of z and converges uniformly, by the same contraction constant, on every compact subdisc. Hence the fixed point is holomorphic. At $z = 0$ it is zero, and comparison of coefficients in (41) identifies its Taylor coefficients with the formal ones.

The bound $|\delta| \leq 1/4$ and Cauchy's estimate on circles of radius $r < 1/(16w)$ imply $|b_n(w)| \leq (1/4)r^{-n}$. Letting r increase to $1/(16w)$ proves (45). Summing the resulting geometric majorant for $n > N$ proves (46).

For the real specialization, $we^{-x} = e^{-w}$, so the estimates give (47). The fixed point is real by uniqueness and conjugation, and $y = w + \delta > 0$. Moreover,

$$\frac{d}{dy}(y + \log y + e^{-y}) = 1 + \frac{1}{y} - e^{-y} > 0 \quad (y > 0).$$

This map tends to $-\infty$ at 0^+ and to $+\infty$ at infinity; therefore its positive inverse is unique and is the solution just constructed. $\square$

The bound is intentionally simple and conservative. It is uniform in the number of retained sectors, not an asymptotic statement with an unspecified constant depending on N . It does *not* solve the repository's general uniformity question for truncated Wright omega or Lambert expansions: w is kept as an exactly inverted core, rather than replaced by its own power-logarithmic truncation. Controlling that second truncation uniformly requires a separate estimate.

9.4 Reproducible numerical checks

The package's script evaluates the first six coefficients at 100 decimal digits and compares truncations with a directly computed root. Selected results are shown below. These floating-point checks illustrate the proven bound; they are not interval-arithmetic certificates and are not used in its proof.

w	N	Observed absolute error	Bound (47)
4	2	$4.5901 \cdot 10^{-6}$	$8.8998 \cdot 10^{-3}$
4	4	$3.2414 \cdot 10^{-9}$	$7.6430 \cdot 10^{-4}$
4	6	$2.9452 \cdot 10^{-12}$	$6.5637 \cdot 10^{-5}$
8	2	$3.9227 \cdot 10^{-11}$	$3.8866 \cdot 10^{-8}$
8	4	$1.1932 \cdot 10^{-17}$	$1.1197 \cdot 10^{-12}$
8	6	$4.6841 \cdot 10^{-24}$	$3.2258 \cdot 10^{-17}$

Values in the last column have been rounded upward at the displayed precision. The full output also includes $w = 5$ and $w = 12$, all orders from one through six, and the associated $x = w + \log w$.

10 Computation, proof certificates, and formalization

10.1 What the support bounds actually make finite

For an input A with $\delta = 1 + v(A) > 0$, the Lagrange formula has a finite *outer* index range below any prescribed power cutoff T . It is enough to retain the positive integers r with

$$r\delta - 1 < T.$$

This is not yet a finite algorithm for every Hahn input. A well-ordered support may contain infinitely many exponents below T , as the example in Section 3 shows. A finite truncation algorithm additionally requires left-finiteness or a symbolic representation that can handle such an infinite initial segment. This distinction is essential when stating complexity claims.

For left-finite input, the following support certificate is sufficient: record the positive set $S = 1 + \text{supp } A$, its least element δ , and the monoid $M(S)$ as a generated object. Products, substitutions, unit inverses, and Newton steps are then authorized by their support inclusions.

One need not form all of $M(S)$ in advance; to work below T , only bounded-length words in the finitely many generators that can contribute need be visited. The finite-factorization lemma guarantees termination. For dense rational or irrational exponent groups, exact comparison of input exponents is an additional representation requirement, not a consequence of the theorem.

Newton's estimate gives a second certificate. To guarantee agreement with the inverse at every exponent strictly below T , it is enough to choose k so that

$$(2^{k+1} - 1)\delta - 1 \geq T.$$

The denominator invariant is independent of this choice: more iterations do not force a finer Puiseux lattice. This separates precision growth from exponent-representation growth.

10.2 Sector-first inversion

For a general admissible map, the efficient organization is:

1. Invert the zero-sector map exactly in the formal power-logarithmic sense, obtaining H_0 with the desired certified power truncation.
2. Transport the positive-sector perturbation to $K = E_+ \circ H_0$ and record $d = \text{ord}_q K$.
3. For sector N , apply (29) only for $1 \leq r \leq \lfloor N/d \rfloor$.
4. Within those finitely many terms, propagate power lower bounds through multiplication and differentiation before truncating.

A uniform power cutoff applied blindly to every intermediate factor is unsafe: multiplication by a negative-valuation coefficient can move an omitted term into the requested range. The bounds from Theorem 8.1 give a particularly simple remedy for a single primitive sector: normalize by $Q = qt^\kappa$, so all resulting coefficient valuations are nonnegative. For arbitrary positive-sector tails, use sector-dependent lower bounds instead of assuming this single straight boundary.

Core normalization is not merely an optimization. It changes a generally nonsummable Lagrange family, such as the constant-shift example of Section 7, into a family with a strict increase of exponential order. A computer algebra system should reject the unnormalized operation unless an additional summation structure has explicitly been supplied.

10.3 The included verification artifact

The accompanying `verify_results.py` has two different roles. Its exact symbolic part checks:

- mixed rational-exponent Lagrange reversion for $A = L + t^{1/2}(L^2 + 1) + t^{2/3}$ modulo terms of valuation at least 4;
- agreement with three Newton steps at that precision, with all exponents remaining in $\frac{1}{6}\mathbb{Z}$;
- the rational formulas for $b_1, \dots, b_6$, their exact pole coefficients and leading powers, and the implicit equation (41) through z^6 .

Its numerical part produces the table data for the analytic example. The exact part uses rational arithmetic and symbolic polynomial/rational simplification in SymPy. The numerical part uses mpmath. A text receipt and a CSV file are supplied so that the reported checks are inspectable without running code. These checks are useful for catching signs, normalizations, and low-order mistakes; they do not replace the quantified proofs in this article.

10.4 A realistic Lean dependency plan

The repository README already distinguishes formalized scale, Neumann, block, and coefficient lemmas from an assembled composition/reversion calculus [2]. The results in this article should be formalized at that same statement-level granularity. The main dependency sequence is:

Layer	Required mathematical interface
Support	Well-ordered support addition; finite fibers; positive-monoid closure; valuation-complete coefficient ring.
Differentiation	A strongly summable derivation with valuation gain; finite polynomial block identities.
Substitution	A Taylor homomorphism with quantitative gain, associativity, chain rule, and the difference estimate.
Reversion	Auxiliary-parameter Lagrange inversion; justified specialization; fixed-point uniqueness; Newton error and support invariants.
Exponential	Ordinary power series over the coefficient ring; sectorwise derivation; kernel and integration; leading-block obstruction.
Core inversion	Group factorization; finite sector coefficient extraction; Cayley boundary and worked rational coefficients.

No new Lean module has been compiled for this article. In particular, a symbolic identity verified by the included Python program is not being promoted to a kernel-checked theorem. The benefit of the present formulation is that both the positive results and the obstruction have small, explicit interfaces suitable for independent formalization.

11 Further research questions

The questions below are proposed directions, not claims of established open status in the entire literature. Some sharpen a repository frontier; others arise from the exact boundary proved here. Their hypotheses are chosen so that a future result can be compared with the present theorems without silently changing the ambient algebra.

Research question 11.1 (The minimal enlargement for quadratic logarithmic shifts). What is the smallest explicitly presented coefficient algebra extending $\mathbb{R}[L]((t^\Gamma))$ that is closed under a specified family of maps $X + P(L) + U$ with $\deg P \geq 2$, and under their inverses? Adjoining $e^{-\mu L^2}$ resolves one immediate missing monomial, but repeated composition also changes its exponent through $\log(X + P(L) + U) = L +$ smaller terms. Can an effective support description track all descendants while keeping coefficient extraction finite?

Theorem 6.1 proves that merely enlarging Γ cannot solve this problem. It does not prove that no larger transseries algebra can do so; standard transseries fields provide much richer ambient structures [3, 5].

Research question 11.2 (Several exponential actions and resonances). For $q_j = e^{-\mu_j X}$, can the admissibility and leading-sector theorems be organized uniformly over a finitely generated action monoid, including rational relations among the μ_j ? In a freely indexed model the conditions $\mu_j a \in \Gamma$ are suggested by the one-generator theorem, but dependent actions identify different formal multi-indices. Determine support and noncancellation hypotheses that make the resulting coefficient formula representation-independent.

Research question 11.3 (Analytic subclasses stable under exact-core inversion). Give coefficient-growth and analytic continuation assumptions on a subclass of $\mathcal{E}_{\Gamma, \mu}$ under which

(29) preserves Borel summability, or a specified resurgent structure. Which singularities can be created when the zero-sector core is inverted, and how do the corresponding Stokes data transform? Formal existence in the product topology alone supplies none of these conclusions.

Research question 11.4 (Two-stage uniform error bounds). In the mixed example, replace the exact w by a truncated power–logarithmic expansion while also truncating the exponential sectors. Find a bound uniform in both truncation orders, with an explicit admissible relation between the orders and X . More generally, formulate conditions on an invertible core that turn the formal support estimate into an analytic residual-to-error certificate with constants uniform over a parameter family.

This would connect the present result to the repository’s `plt:rmk:ext-open-uniformity`, without confusing a bound for the exponential tail with a bound for the core’s own asymptotic expansion.

Research question 11.5 (Beyond Archimedean exponent groups). When the exponent group is an arbitrary ordered abelian group, what replaces the real-valued hypothesis used in the coefficientwise Lagrange bound and the sequential fixed-point proof? A positive gain δ need not have multiples cofinal in the whole group. Identify conditions under which strong Hahn summability still suffices, and distinguish genuine failures of existence from failures of sequential convergence in a chosen topology.

The current article deliberately restricts to $\Gamma \subseteq \mathbb{R}$; no non-Archimedean extension is implicit in its proofs.

Research question 11.6 (Exact boundaries for several primitive sectors). If $E_+ = \sum_{j=1}^d q^j f_j$, determine the smallest possible power valuation at each output sector and classify when its leading coefficient cancels. A finite minimization over sector decompositions predicts a lower bound, but the exact boundary requires algebraic control of all minimizing terms. Seek hypotheses under which an analogue of the nonvanishing Cayley law can be proved.

Research question 11.7 (Realization beyond Puiseux supports). Which left-finite rational-exponent power–logarithmic series outside every fixed Puiseux lattice admit compatible realizations in a Hardy field, with formal differentiation, composition, and inversion agreeing with those of the representing germs? The support theorems here separate the formal problem from this analytic realization problem; they do not select a canonical germ for an arbitrary formal series.

This is a sharpened continuation of `plt:rmk:ext-open-realization`, not a consequence of denominator stability for a fixed input lattice.

Research question 11.8 (Sharp complex transition near the core critical point). For the mixed example, can the deepest-pole expression (44) be promoted to a uniform scaling approximation near $w = -1$, with both error bounds and the actual nearby branch points located? The rational pole hierarchy is exact, but its formal leading-pole sum is not yet a uniform two-variable approximation to the complete inverse.

Research question 11.9 (Proof-producing support and precision inference). Can a symbolic engine infer the support monoid, admissible exponent lattice, and required intermediate precisions automatically, then emit small independently checked certificates? A useful first target is a Lean-checked implementation of the fixed-Puiseux and finitely generated action cases, with explicit rejection of nonsummable unnormalized substitutions.

12 Conclusion

The dense-exponent reversion problem does not require replacing the positive-integer Lagrange index: the normalized valuation gain supplies exactly the finiteness that density was thought to obstruct. Newton iteration preserves more than a denominator bound—it preserves the positive support monoid generated by the input. These conclusions settle the stated dense-Hahn and Puiseux-iteration questions in their specified real-exponent settings.

The one-exponential problem has a different answer. The polynomial-logarithmic coefficient ring cannot support arbitrary logarithmic shifts once $q = e^{-\mu X}$ must transform differentially correctly. Among sublinear Hahn base displacements, the exact surviving class consists of affine logarithmic shifts with $\mu a \in \Gamma$, positive power-order corrections, and arbitrary positive exponential sectors. Within that class, block operations, integration, composition, and inversion are rigorous and explicit. Exact-core normalization is the mechanism that restores summable Lagrange inversion.

For a single primitive exponential perturbation, every inverse sector has an exact leading power and a nonzero Cayley coefficient. The mixed inverse of $X + \log X + e^{-X}$ shows how this formal statement can meet analysis: rational coefficient functions, sharp pole orders, and a uniform geometric remainder bound hold simultaneously. The remaining issues—richer exponent alphabets, multi-action cancellations, analytic realization and resurgence, and complete formal verification—are now separated from the claims actually proved.

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